Supplementary Material to "Logic-Based Benders decomposition for additive manufacturing scheduling with sequence-dependent setup times" by Wenjing Ma, Jian Chen, Yaowen Sang, Yingqian Zhang, Yaoxin Wu

S.1 Data generation

We selected five different types of machines and five representative material types (Ti64, AlSi10Mg, Stainless Steel 316L(316L), Nickel Alloy IN718 (IN718), and Maraging Steel MS1 (MS1)). These materials are compatible with the selected machines, allowing us to evaluate their respective manufacturing times across different machine configurations. For each material, we identified representative layer thickness values and obtained corresponding unit volume build times (VT_{iu}) and unit height build times (HT_{iu}). Table 7-10 below show the machine specifications (the maximum build area MA_i and the maximum build height MH_i), layer thickness, unit volume build time (VT_{iu}), and unit height build time (HT_{iu}) for each material across different machines.

Table 7 Machine specifications

Machine	Build volume (cm^3)	$MA_i(cm^2)$	$MH_i(cm)$
EOS M290	25 × 25 × 32.5	625	32.5
EOS M290-2	25 × 25 × 32.5	625	32.5
EOS 300-4	30 × 30 × 43.5	900	43.5
EOS M400	40 × 40 × 40	1600	40
EOS M400-4	40 × 40 × 40	1600	40

Table 7 presents the specifications of the machines used in this study. The "Build Volume" refers to the maximum printing space in cubic centimeters, while MA_i is the maximum printable area (in cm²) and MH_i represents the maximum build height (in cm) for each machine.

Table 8 Layer thickness ($lt_{iu}(\mu m)$) for different materials and machines

Machine	Ti64	AlSi10Mg	316L	IN718	MS1
EOS M290	30	30	40	40	40
EOS M290-2	60	60	80	80	50
EOS 300-4	60	60	80	40	50
EOS M400	30	90	80	40	40
EOS M400-4	60	30	40	40	40

Table 9 The build time per unit volume ($VT_{iu}(h/cm^3)$)

Machine	Ti64	AlSi10Mg	316L	IN718	MS1
EOS M290	0.055556	0.054466	0.075075	0.066139	0.066139
EOS M290-2	0.030864	0.026455	0.033069	0.033875	0.025253
EOS M300-4	0.007716	0.006614	0.008267	0.016535	0.012626
EOS M400	0.030864	0.009992	0.033069	0.066139	0.053419
EOS M400-4	0.007716	0.009384	0.018769	0.013355	0.016535

The corresponding values for build time per unit height (HT_{iu}) are computed using the layer thickness and recoating time per layer. For each machine, the recoating time per layer t_i^{recoat} is determined using a travel-time model $t_i^{\text{recoat}} = 2W_i/v + c$, where W_i is the recoater travel distance (approximated by the build width of the machine), v is the recoater speed (set to 250 mm/s), and c is the fixed overhead time (set to 0.5 seconds). The build time per unit height is then computed as $HT_{iu} = 10^4 t_i^{\text{recoat}} / 3600 lt_{iu}$ (h/cm), where lt_{iu} is the layer thickness of material u on machine i. Table 10 presents the unit height build times (HT_{iu}) for each material on the selected machines.

Table 10 The build time per unit height ($HT_{iu}(h/cm)$)

Machine	Ti64	AlSi10Mg	316L	IN718	MS1
EOS M290	0.231481	0.231481	0.173611	0.173611	0.173611
EOS M290-2	0.115741	0.115741	0.086806	0.086806	0.138889
EOS M300-4	0.134259	0.134259	0.100694	0.201389	0.161111
EOS M400	0.342593	0.124198	0.128472	0.256944	0.256944
EOS M400-4	0.171296	0.342593	0.256944	0.256944	0.256944

Table 11 presents the specifications of the parts. The part-related parameters include h_r, the height of part r, a_r, the area of part r, and v_r, the volume of part r, all of which are uniformly distributed within the corresponding intervals.

Table 11 Summary of part specifications

Parameter	Value
$h_r(cm)$	$\text{U}[5, 40]$
$a_r(cm^2)$	$\text{U}[50, 800]$
$v_r(cm^3)$	$\text{U}[25, 0.1 \cdot a_r \cdot h_r]$

To better reflect the practical characteristics of material switching in DLMS, we define the sequence-dependent setup time as a machine- and material-pair-dependent parameter. We construct a base material-switching matrix, denoted as $\bar{s}_{uv}$, that captures the asymmetric cleaning burden associated with switching from material u to material v. This approach reflects the fact that the time required for material transitions is not symmetric. For example, the transition from other materials to Ti64 requires more time due to the stricter cleaning protocols and higher contamination sensitivity of titanium alloys. Similarly, same-material transitions (e.g., from 316L to 316L) have lower setup times due to the simplicity of the process, as no cross-contamination cleaning is required.

Table 12 The base setup time matrix $\bar{s}_{uv}$ (in hours)

Material	Ti64	AlSi10Mg	316L	IN718	MS1
Ti64	1	3	3.5	4	4
AlSi10Mg	4	1	2.5	3	3.5
316L	4.5	2.5	1	2	2
IN718	5	3	2	1	2
MS1	5	3.5	2	2	1

The final setup time for a given machine i and material transition u to v is then calculated as: $set_{iuv}^{2} = \alpha_i \bar{s}_{uv}$, where α_i is a machine-specific scaling factor that accounts for differences in machine chamber size, cleaning complexity, and the number of lasers used in the system. Since directly reported material-to-material setup-time matrices for DLMS systems are rarely available in the literature, the values in the matrix are constructed based on practice-oriented considerations. These considerations include factors such as: i) cross-contamination risk (where transitions to materials like Ti64 are given higher values due to the increased risk of contamination); ii) material sensitivity (with titanium alloys requiring more careful handling and cleaning); iii) cleaning burden (with transitions between steels typically requiring less cleaning effort compared to metal-to-titanium transitions). The base material-switching matrix $\bar{s}_{uv}$ is constructed based on these practical considerations and is presented in Table 12. Additionally, we define the setup time set_{iu}^{1} for starting the first batch with material u on machine i, which represents the time required to transition from no material to the target material. The base initial setup time vector for the first batch of each material is defined as $s_u = [1,1,1,0.5,0.5]$. The final setup time for the first batch is then calculated as: $set_{iu}^{1} = \alpha_i s_u$. For the five machines considered in this study, the machine-specific scaling factor α_i is defined as $\alpha_i = [1,1,1.2,1.5,1.5]$.

S.2 Detailed Computational Results

In the following, we report the detailed results from our computational study.

Table 1 Detailed results of three MIP models

Instances	MIP1			MIP2			MIP3		
	Obj	Gap	Time	Obj	Gap	Time	Obj	Gap	Time
I2_R10_M2-1	35.64	91.58	3600	35.64	0	12.03	35.64	47.59	3600
I2_R10_M2-2	224.98	9.70	3600	224.98	0	0.99	224.98	1.52	3600
I2_R10_M2-3	134.61	0	150.61	134.61	0	1.74	134.61	8.36	3600
I2_R10_M2-4	60.80	91.54	3600	60.80	0	223.82	60.80	53.14	3600
I2_R10_M2-5	72.95	95.06	3600	72.95	0	4.50	72.95	25.97	3600
I2_R10_M3-1	75.55	84.12	3600	75.55	0	2.65	75.55	23.34	3600
I2_R10_M3-2	51.77	68.08	3600	51.77	0	2.79	51.77	17.38	3600
I2_R10_M3-3	69.98	77.48	3600	69.98	0	13.37	69.98	12.00	3600
I2_R10_M3-4	93.49	98.40	3600	80.99	0	243.19	80.99	65.38	3600
I2_R10_M3-5	127.98	78.30	3600	127.31	0	3.65	127.98	14.26	3600
I2_R20_M2-1	233.37	93.49	3600	233.37	0	336.23	233.37	21.71	3600
I2_R20_M2-2	167.39	99.10	3600	167.39	8.29	3600	167.39	41.53	3600
I2_R20_M2-3	97.32	98.97	3600	97.18	9.48	3600	97.15	97.63	3600
I2_R20_M2-4	146.55	92.62	3600	146.55	0	152.07	146.55	30.73	3600
I2_R20_M2-5	121.71	98.77	3600	121.71	7.11	3600	121.68	95.35	3600
I2_R20_M3-1	70.84	98.72	3600	70.84	31.73	3600	70.84	98.75	3600
I2_R20_M3-2	96.06	97.44	3600	96.06	0	1159.02	96.06	34.90	3600
I2_R20_M3-3	87.67	96.19	3600	87.67	0	2411.27	87.67	29.59	3600
I2_R20_M3-4	119.02	98.74	3600	119.02	11.05	3600	119.02	45.83	3600
I2_R20_M3-5	116.02	99.18	3600	116.02	8.90	3600	116.02	63.45	3600
I5_R10_M2-1	21.03	92.87	3600	21.03	0	5.45	21.03	0	745.41
I5_R10_M2-2	46.09	96.87	3600	46.09	0	481.96	46.09	59.45	3600
I5_R10_M2-3	28.31	97.35	3600	28.31	0	2.30	28.31	0	19.42
I5_R10_M2-4	57.51	35.56	3600	57.51	0	2.97	57.51	13.86	3600
I5_R10_M2-5	18.50	59.46	3600	18.50	0	316.16	18.50	8.27	3600
I5_R10_M3-1	23.44	93.60	3600	23.44	0	2.12	23.44	0	25.14
I5_R10_M3-2	28.15	94.67	3600	28.15	0	3.46	28.15	0	151.59
I5_R10_M3-3	61.21	97.55	3600	61.21	0	9.41	61.21	15.61	3600
I5_R10_M3-4	34.91	98.19	3600	34.91	0	136.02	34.91	25.92	3600
I5_R10_M3-5	17.28	96.53	3600	17.28	0	616.27	17.28	59.62	3600

I5_R20_M2-1	31.77	97.55	3600	31.33	21.50	3600	31.33	97.60	3600
I5_R20_M2-2	49.54	100	3600	49.54	2.46	3600	49.54	29.86	3600
I5_R20_M2-3	46.76	98.40	3600	46.76	0	33.85	46.71	98.39	3600
I5_R20_M2-4	40.31	99.07	3600	37.83	19.72	3600	37.83	98.01	3600
I5_R20_M2-5	44.95	96.66	3600	44.95	0	2180.09	44.95	35.36	3600
I5_R20_M3-1	39.81	96.99	3600	39.81	0	2712.67	39.81	64.52	3600
I5_R20_M3-2	70.19	100	3600	70.19	0	2411.91	70.19	46.64	3600
I5_R20_M3-3	45.02	96.67	3600	45.02	0	310.46	45.02	44.69	3600
I5_R20_M3-4	36.03	100	3600	36.03	17.98	3600	35.99	69.80	3600
I5_R20_M3-5	52.33	100	3600	52.13	7.68	3600	52.13	99.04	3600

Table 2 Detailed results of three MIPV models

Instances	MIP1V			MIP2V			MIP3V		
	Obj	Gap	Time	Obj	Gap	Time	Obj	Gap	Time
I2_R10_M2-1	35.64	0	6.16	35.64	0	14.78	35.64	0	16.56
I2_R10_M2-2	224.98	0	1.08	224.98	0	0.46	224.98	0	0.89
I2_R10_M2-3	134.61	0	376.42	134.61	0	3.11	134.61	0	2.71
I2_R10_M2-4	60.80	0	1022.39	60.80	0	21.97	60.80	0	110.32
I2_R10_M2-5	72.95	0	7.78	72.95	0	8.40	72.95	0	2.81
I2_R10_M3-1	75.55	10.25	3600	75.55	0	1.31	75.55	0	1.83
I2_R10_M3-2	51.77	0	3.48	51.77	0	1.43	51.77	0	1.59
I2_R10_M3-3	69.98	0	29.78	69.98	0	7.20	69.98	0	9.08
I2_R10_M3-4	93.49	0	13.15	80.99	0	6.82	80.99	0	5.28
I2_R10_M3-5	127.98	5.56	3600	120.31	0	3.11	120.31	0	6.18
I2_R20_M2-1	233.37	2.82	3600	233.37	0	336.23	233.37	5.78	3600
I2_R20_M2-2	167.39	3.11	3600	167.39	8.29	3600	167.39	23.11	3600
I2_R20_M2-3	97.17	2.28	3600	97.18	8.87	3600	97.15	35.28	3600
I2_R20_M2-4	146.55	3.65	3600	146.55	0	152.07	146.55	9.33	3600
I2_R20_M2-5	121.71	0	15.99	121.71	6.74	3600	121.68	36.72	3600
I2_R20_M3-1	70.84	1.29	3600	70.84	23.98	3600	70.84	46.62	3600
I2_R20_M3-2	96.06	5.09	3600	96.06	0	1826.72	96.06	15.84	3600
I2_R20_M3-3	87.67	9.97	3600	87.67	0	1713.63	87.67	21.35	3600
I2_R20_M3-4	119.02	7.80	3600	119.02	10.86	3600	119.02	21.75	3600
I2_R20_M3-5	116.02	2.98	3600	116.02	7.93	3600	116.02	13.62	3600
I5_R10_M2-1	21.03	0	2.92	21.03	0	5.45	21.03	0	1.78
I5_R10_M2-2	46.09	0	3.10	46.09	0	481.96	46.09	0	6.71
I5_R10_M2-3	28.31	0	2.03	28.31	0	2.30	28.31	0	1.87
I5_R10_M2-4	57.51	0	3.12	57.51	0	2.97	57.51	0	0.94
I5_R10_M2-5	18.50	0	4.10	18.50	0	316.16	18.50	0	29.27
I5_R10_M3-1	23.44	0	2.88	23.44	0	2.12	23.44	0	1.47

I5_R10_M3-2	28.15	0	1.99	28.15	0	3.46	28.15	0	2.85
I5_R10_M3-3	61.21	0	18.32	61.21	0	9.41	61.21	0	2.24
I5_R10_M3-4	34.91	0	11.99	34.91	0	136.02	34.91	0	6.65
I5_R10_M3-5	17.28	0	4.61	17.28	0	616.27	17.28	0	7.30
I5_R20_M2-1	31.77	21.50	3600	31.33	0	812.78	31.33	57.17	3600
I5_R20_M2-2	49.54	2.46	3600	49.54	0	212.87	49.54	18.82	3600
I5_R20_M2-3	46.76	0	33.85	46.76	0	44.08	46.71	0	1081.24
I5_R20_M2-4	37.83	19.72	3600	37.83	0	48.40	37.83	27.21	3600
I5_R20_M2-5	44.95	0	62.51	44.95	0	68.05	44.95	0	766.67
I5_R20_M3-1	39.81	0	2712.67	39.81	0	140.13	39.81	9.20	3600
I5_R20_M3-2	70.19	0	2411.91	70.19	0	266.58	70.19	16.66	3600
I5_R20_M3-3	45.02	0	310.46	45.02	1.47	3600	45.02	24.70	3600
I5_R20_M3-4	36.03	17.98	3600	36.03	0	998.72	35.99	35.14	3600
I5_R20_M3-5	52.13	7.68	3600	52.13	0	630.86	52.13	18.70	3600

Table 3 Detailed results of MIP2V and LBBD of small-sized instances

Instances	MIP2V			LBBD		
	Obj	Gap	Time	Obj	Gap	Time
I2_R10_M2-1	35.64	0	14.78	35.64	0	1.15
I2_R10_M2-2	224.98	0	0.46	224.98	0	0.12
I2_R10_M2-3	134.61	0	3.11	134.61	0	1.36
I2_R10_M2-4	60.80	0	21.97	60.80	0	2.42
I2_R10_M2-5	72.95	0	8.40	72.95	0	0.40
I2_R10_M3-1	75.55	0	1.31	75.55	0	0.56
I2_R10_M3-2	51.77	0	1.43	51.77	0	1.06
I2_R10_M3-3	69.98	0	7.20	69.98	0	0.14
I2_R10_M3-4	80.99	0	6.82	80.99	0	1.33
I2_R10_M3-5	120.31	0	3.11	120.31	0	0.81
I2_R20_M2-1	233.37	0	336.23	233.37	0	2.15
I2_R20_M2-2	167.39	8.29	3600	167.22	0	23.18
I2_R20_M2-3	97.18	8.87	3600	97.18	2.43	3600
I2_R20_M2-4	146.55	0	152.07	146.55	0	9.26
I2_R20_M2-5	121.71	6.74	3600	121.71	0	1252.54
I2_R20_M3-1	70.84	23.98	3600	70.84	0	12.92
I2_R20_M3-2	96.06	0	1826.72	96.06	0	20.80
I2_R20_M3-3	87.67	0	1713.63	87.67	0	5.69
I2_R20_M3-4	119.02	10.86	3600	119.02	0	16.48
I2_R20_M3-5	116.02	7.93	3600	116.02	0	101.58

I3_R10_M2-1	20.14	0	2.89	20.14	0	1.52
I3_R10_M2-2	50.61	0	2.77	50.61	0	0.32
I3_R10_M2-3	35.84	0	35.50	35.84	0	1.65
I3_R10_M2-4	31.20	0	10.06	31.21	0	1.30
I3_R10_M2-5	39.38	0	0.06	39.38	0	0.10
I3_R10_M3-1	32.62	0	2.54	32.62	0	0.93
I3_R10_M3-2	40.29	0	0.34	40.29	0	0.12
I3_R10_M3-3	27.71	0	46.47	27.71	0	2.18
I3_R10_M3-4	47.91	0	2.71	47.91	0	1.15
I3_R10_M3-5	17.08	0	6.69	17.08	0	0.74
I3_R20_M2-1	43.56	12.85	3600	43.56	0	11.48
I3_R20_M2-2	73.40	16.24	3600	73.40	0	46.56
I3_R20_M2-3	56.76	20.74	3600	56.76	3.11	3600
I3_R20_M2-4	54.95	23.97	3600	54.95	0	11.47
I3_R20_M2-5	70.71	0	9.05	70.71	0	7.62
I3_R20_M3-1	55.09	29.78	3600	55.09	0	17.74
I3_R20_M3-2	76.97	40.30	3600	76.77	0	12.58
I3_R20_M3-3	45.45	0	1014.74	45.45	0	5.89
I3_R20_M3-4	68.81	40.78	3600	68.81	0	479.05
I3_R20_M3-5	66.76	25.01	3600	66.76	0	24.19
I5_R10_M2-1	21.03	0	5.45	21.03	0	0.90
I5_R10_M2-2	46.09	0	481.96	46.09	0	0.85
I5_R10_M2-3	28.31	0	2.30	28.31	0	0.36
I5_R10_M2-4	57.51	0	2.97	57.51	0	0.08
I5_R10_M2-5	18.50	0	316.16	18.50	0	1.23
I5_R10_M3-1	23.44	0	2.12	23.44	0	0.19
I5_R10_M3-2	28.15	0	3.46	28.15	0	0.78
I5_R10_M3-3	61.21	0	9.41	61.21	0	1.26
I5_R10_M3-4	34.91	0	136.02	34.91	0	0.98
I5_R10_M3-5	17.28	0	616.27	17.28	0	3.84
I5_R20_M2-1	31.33	21.50	3600	31.33	0	27.73
I5_R20_M2-2	49.54	2.46	3600	49.54	0	16.35
I5_R20_M2-3	46.76	0	33.85	46.76	0	1.58
I5_R20_M2-4	37.83	19.72	3600	37.83	0	14.09
I5_R20_M2-5	44.95	0	2180.09	44.95	0	3.44
I5_R20_M3-1	39.81	0	2712.67	39.81	0	5.05
I5_R20_M3-2	70.19	0	2411.91	70.19	0	6.01
I5_R20_M3-3	45.02	0	310.46	45.02	0	3.29
I5_R20_M3-4	36.03	17.98	3600	36.03	0	108.34
I5_R20_M3-5	52.13	7.68	3600	52.13	0	17.99

Table 4 Detailed results of MIP2V and LBBD of medium-sized instances

	MIP2V			LBBD		
	Obj	Gap	Time	Obj	Gap	Time
I2_R30_M2-1	105.36	14.48	3600	104.32	0	197.80
I2_R30_M2-2	193.78	25.82	3600	193.28	0	2506.76
I2_R30_M2-3	83.08	22.22	3600	81.53	3.86	3600
I2_R30_M2-4	92.02	10.75	3600	89.38	4.50	3600
I2_R30_M2-5	108.84	13.72	3600	107.85	2.51	3600
I2_R30_M3-1	98.15	27.54	3600	97.47	0	3591.18
I2_R30_M3-2	198.24	30.06	3600	198.24	0	11.41
I2_R30_M3-3	92.02	15.00	3600	91.51	4.55	3600
I2_R30_M3-4	133.47	33.86	3600	133.03	5.17	3600
I2_R30_M3-5	328.80	12.85	3600	328.80	0.90	3600
I2_R40_M2-1	137.79	17.62	3600	136.23	4.63	3600
I2_R40_M2-2	261.68	33.09	3600	259.43	1.31	3600
I2_R40_M2-3	68.56	3.24	3600	68.19	3.52	3600
I2_R40_M2-4	112.09	12.73	3600	110.85	6.97	3600
I2_R40_M2-5	394.36	1.47	3600	393.08	0.54	3600
I2_R40_M3-1	169.46	27.56	3600	165.20	2.56	3600
I2_R40_M3-2	160.94	36.64	3600	159.20	1.83	3600
I2_R40_M3-3	116.61	25.47	3600	113.92	8.53	3600
I2_R40_M3-4	243.36	5.33	3600	236.35	2.93	3600
I2_R40_M3-5	171.59	35.11	3600	171.35	3.75	3600
I3_R30_M2-1	69.98	22.33	3600	69.98	0	373.96
I3_R30_M2-2	119.08	14.16	3600	118.78	0	3013.63
I3_R30_M2-3	79.70	19.13	3600	78.45	4.32	3600
I3_R30_M2-4	61.84	16.55	3600	61.67	4.47	3600
I3_R30_M2-5	144.07	15.07	3600	142.75	0.77	3600
I3_R30_M3-1	67.30	33.05	3600	66.84	2.73	3600
I3_R30_M3-2	90.54	18.72	3600	88.46	4.53	3600
I3_R30_M3-3	88.17	19.42	3600	87.48	3.83	3600
I3_R30_M3-4	126.77	15.59	3600	126.77	0	487.16
I3_R30_M3-5	289.49	7.24	3600	289.49	0	17.24
I3_R40_M2-1	112.59	30.00	3600	110.98	5.09	3600
I3_R40_M2-2	139.59	23.49	3600	134.81	5.22	3600
I3_R40_M2-3	48.31	12.13	3600	42.34	0	261.75
I3_R40_M2-4	79.39	11.88	3600	77.53	5.05	3600
I3_R40_M2-5	371.08	7.24	3600	371.08	0	29.20

I3_R40_M3-1	335.23	16.15	3600	335.23	0	44.93
I3_R40_M3-2	75.03	26.04	3600	72.60	6.16	3600
I3_R40_M3-3	100.95	14.86	3600	100.95	0	5.55
I3_R40_M3-4	98.54	33.76	3600	97.93	10.86	3600
I3_R40_M3-5	120.19	33.36	3600	117.32	1.63	3600
I5_R30_M2-1	47.41	14.40	3600	47.41	0	488.01
I5_R30_M2-2	81.98	9.42	3600	81.38	0	1087.77
I5_R30_M2-3	57.12	0	1461.39	57.12	0	4.80
I5_R30_M2-4	49.81	21.38	3600	49.93	7.84	3600
I5_R30_M2-5	72.63	40.57	3600	72.26	2.26	3600
I5_R30_M3-1	52.91	36.67	3600	51.78	3.32	3600
I5_R30_M3-2	78.93	42.85	3600	76.27	4.19	3600
I5_R30_M3-3	53.02	16.13	3600	52.92	4.19	3600
I5_R30_M3-4	50.65	42.82	3600	48.17	10.07	3600
I5_R30_M3-5	176.78	17.89	3600	176.78	0	21.58
I5_R40_M2-1	81.90	41.47	3600	76.12	27.99	3600
I5_R40_M2-2	74.62	19.79	3600	72.69	18.24	3600
I5_R40_M2-3	128.16	10.46	3600	128.16	0	8.28
I5_R40_M2-4	72.70	32.49	3600	71.30	20.76	3600
I5_R40_M2-5	277.99	7.41	3600	277.99	0	31.05
I5_R40_M3-1	129.60	43.53	3600	128.93	6.46	3600
I5_R40_M3-2	63.37	32.56	3600	60.92	5.54	3600
I5_R40_M3-3	104.82	14.31	3600	104.82	0	8.89
I5_R40_M3-4	63.46	36.13	3600	61.58	7.02	3600
I5_R40_M3-5	86.11	23.86	3600	84.77	0	7.48

Table 5 Detailed results of MIP2V and LBBD of large-sized instances

	MIP2V			**LBBD**		
	Obj	Gap	Time	Obj	Gap	Time
I5_R50_M3-1	111.78	21.10	3600	106.25	4.93	3600
I5_R50_M5-2	89.89	33.76	3600	89.30	12.00	3600
I5_R50_M3-3	47.59	21.96	3600	44.65	16.76	3600
I5_R50_M3-4	124.47	27.41	3600	120.58	10.19	3600
I5_R50_M3-5	83.21	39.22	3600	79.99	4.80	3600
I5_R50_M5-1	93.11	46.54	3600	86.01	11.82	3600
I5_R50_M5-2	96.36	37.49	3600	96.36	0	554.85
I5_R50_M5-3	74.55	39.51	3600	72.30	15.17	3600
I5_R50_M5-4	123.36	51.52	3600	117.90	4.34	3600
I5_R50_M5-5	110.35	43.78	3600	105.01	5.27	3600
I5_R80_M3-1	183.55	31.17	3600	174.85	5.92	3600

I5_R80_M3-2	300.30	49.54	3600	289.16	3.02	3600
I5_R80_M3-3	134.13	41.12	3600	127.49	16.23	3600
I5_R80_M3-4	152.97	40.27	3600	149.85	9.60	3600
I5_R80_M3-5	120.74	28.40	3600	115.87	11.37	3600
I5_R80_M5-1	371.48	59.76	3600	356.71	10.74	3600
I5_R80_M5-2	133.93	28.71	3600	120.44	17.03	3600
I5_R80_M5-3	131.94	31.88	3600	123.10	15.87	3600
I5_R80_M5-4	161.77	28.63	3600	153.24	12.36	3600
I5_R80_M5-5	169.15	39.44	3600	162.03	11.95	3600
I5_R100_M3-1	224.80	14.01	3600	222.55	1.65	3600
I5_R100_M3-2	637.40	68.20	3600	631.61	1.67	3600
I5_R100_M3-3	192.94	51.43	3600	187.69	3.11	3600
I5_R100_M3-4	175.82	31.28	3600	159.17	13.05	3600
I5_R100_M3-5	207.23	35.69	3600	205.31	12.13	3600
I5_R100_M5-1	208.43	44.01	3600	180.51	12.96	3600
I5_R100_M5-2	174.66	37.88	3600	160.22	15.34	3600
I5_R100_M5-3	184.81	42.32	3600	177.27	14.85	3600
I5_R100_M5-4	266.56	50.94	3600	241.61	19.04	3600
I5_R100_M5-5	175.94	30.02	3600	164.88	16.32	3600
I10_R50_M3-1	44.87	47.48	3600	45.08	11.61	3600
I10_R50_M3-2	47.44	30.85	3600	46.36	16.40	3600
I10_R50_M3-3	25.47	29.92	3600	24.64	11.58	3600
I10_R50_M3-4	44.38	16.20	3600	44.38	0.00	2871.19
I10_R50_M3-5	37.52	31.38	3600	37.35	5.98	3600
I10_R50_M5-1	57.31	49.79	3600	51.44	12.28	3600
I10_R50_M5-2	51.39	38.36	3600	51.48	18.49	3600
I10_R50_M5-3	55.23	49.41	3600	52.38	12.55	3600
I10_R50_M5-4	55.15	42.18	3600	55.15	12.82	3600
I10_R50_M5-5	51.10	46.61	3600	50.79	13.67	3600
I10_R80_M3-1	81.63	42.29	3600	75.82	12.19	3600
I10_R80_M3-2	82.63	47.54	3600	76.99	10.82	3600
I10_R80_M3-3	70.25	49.17	3600	67.90	16.94	3600
I10_R80_M3-4	75.03	44.13	3600	68.13	16.80	3600
I10_R80_M3-5	67.56	39.79.21	3600	62.67	9.82	3600
I10_R80_M5-1	105.30	25.49	3600	94.91	5.66	3600
I10_R80_M5-2	65.55	33.27	3600	59.98	18.80	3600
I10_R80_M5-3	85.61	49.79	3600	78.92	20.50	3600
I10_R80_M5-4	92.34	46.16	3600	89.84	9.17	3600
I10_R80_M5-5	63.83	46.55	3600	59.11	23.55	3600
I10_R100_M3-1	112.55	53.78	3600	111.05	1.74	3600

I10_R100_M3-2	100.14	46.97	3600	95.12	11.25	3600
I10_R100_M3-3	76.31	38.48	3600	71.56	19.51	3600
I10_R100_M3-4	105.63	44.90	3600	94.54	22.37	3600
I10_R100_M3-5	80.38	35.83	3600	74.98	17.23	3600
I10_R100_M5-1	91.09	38.33	3600	85.25	21.33	3600
I10_R100_M5-2	85.51	38.71	3600	74.36	22.26	3600
I10_R100_M5-3	107.07	45.04	3600	97.87	17.00	3600
I10_R100_M5-4	156.52	58.87	3600	151.51	9.44	3600
I10_R100_M5-5	93.09	47.21	3600	85.43	24.37	3600